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DISHWASHER SECURING SYSTEMS

Abstract

Embodiments of a dishwasher securing system are described herein. The dishwasher securing system may include a dishwasher, the dishwasher having a housing comprising a top face, a bottom face, and one or more side faces. The system may further include a threaded housing, the threaded housing coupled to the top face of the dishwasher, the threaded housing having a plurality of threaded holes. A first threaded leg may extend from the threaded housing, the first threaded leg having one or more threads configured to engage at least one of the plurality of threaded holes of the threaded housing. A second threaded leg may extend from the threaded housing, the second threaded leg having one or more threads configured to engage at least one of the plurality of threaded holes of the threaded housing.

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Background/Summary

BACKGROUND

[0001] The present embodiments relate to systems for securing dishwashers, with particular embodiments shown for size adjustable systems for securing dishwashers within a cabinet opening or other similar opening.

[0002] Typical dishwasher securing systems may be configured to use brackets and mechanical fasteners to secure the dishwasher within a cabinet opening. This may create holes in the cabinets when fasteners are driven therethrough. If multiple dishwashers are installed in the same cabinet opening over time, the cabinets may be weakened by multiple holes being formed therein. Further, the brackets may not be size adjustable, which may require different length brackets to be used to accommodate different sized cabinet openings. Thus, there is a need for a dishwasher securing system that is size adjustable and that may secure the dishwasher without requiring holes to be formed in the cabinet opening.

SUMMARY

[0003] The herein-described embodiments address these and other problems associated with the art by providing systems for securing dishwashers within a cabinet opening or other openings. In various embodiments, a dishwasher securing system may include a dishwasher, the dishwasher having a housing comprising a top face, a bottom face, and one or more side faces. The system may further include a threaded housing, the threaded housing coupled to the top face of the dishwasher, and the threaded housing having a plurality of threaded holes. A first threaded leg may extend from the threaded housing, the first threaded leg having one or more threads configured to engage at least one of the plurality of threaded holes of the threaded housing. A second threaded leg may extend from the threaded housing, the second threaded leg having one or more threads configured to engage at least one of the plurality of threaded holes of the threaded housing.

[0004] In addition, in some embodiments a dishwasher securing system may include a dishwasher, the dishwasher having a housing having a top face, a bottom face, and a plurality of side faces. The system may further include one or more threaded extensions coupled to one of the plurality of side faces. Each of the one or more threaded extensions may be configured to be threaded outward relative to one of the plurality of side faces of the dishwasher.

[0005] In addition, in some embodiments a dishwasher securing system may include a dishwasher, the dishwasher having a housing having a top face, a bottom face, and a plurality of side faces. The system may include a plurality of racks, each of the plurality of racks coupled to the top face of the dishwasher having a plurality of grooved teeth. One or more arms may be coupled to each of the plurality of racks. The system may include a pinion. The pinion may include a plurality of gear teeth. The pinion may be configured to engage the plurality of grooved teeth of the plurality of racks. Each of the plurality of racks may be configured to extend outwardly when engaged by the pinion.

[0006] In addition, in some embodiments a dishwasher securing system may include a dishwasher the dishwasher having a housing comprising a top face, a bottom face, and one or more side faces. The system may further include a center pulley, the center pulley rotatably coupled to the top face of the dishwasher, wherein the center pulley comprises a lower groove and an upper groove. The system may further include a first extending arm comprising a first outer pulley, the first extending arm rotatably coupled to the top face of the dishwasher about the first outer pulley, wherein the first outer pulley comprises a first groove. The system may further include a second extending arm comprising a second outer pulley, the second extending arm rotatably coupled to the top face of the dishwasher about the second outer pulley, wherein the second outer pulley comprises a second groove. The system may further include a first cable, the first cable forming a continuous loop and coupled to the first groove of the first outer pulley and one of the upper groove or the lower groove of the center pulley. The system may further include a second cable, the second cable forming a continuous loop and coupled to the second groove of the second outer pulley and one of the upper groove or the lower groove of the center pulley, wherein the first cable and second cable are each coupled to a different groove of the center pulley.

[0007] In addition, in some embodiments a dishwasher securing system may include a dishwasher the dishwasher having a housing comprising a top face, a bottom face, and one or more side faces. The system may further include a first pin disposed through a first pin hole formed in one of the one or more side faces of the dishwasher. The system may further include a second pin disposed through a second pin hole formed in one of the one or more side faces of the dishwasher. The system may further include a first spring configured to bias the first pin outwardly from one of the one or more side faces of the dishwasher. The system may further include a second spring configured to bias the second pin outwardly from one of the one or more side faces of the dishwasher.

[0008] These and other advantages and features are set forth in the claims annexed hereto and forming a further part hereof. However, for a better understanding of the invention, and of the advantages and objectives attained through its use, reference should be made to the figures, and to the accompanying descriptive matter, in which there is described example embodiments of the invention. This summary is merely provided to introduce a selection of concepts that are further described below in the detailed description, and is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used in any way to the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is an isometric view of a dishwasher disposed within a cabinet opening, consistent with some embodiments of the dishwasher securing systems.

[0010] FIG. **2** is a front view of a dishwasher securing system, consistent with some embodiments of the dishwasher securing system.

[0011] FIG. **3** is a top view of a dishwasher securing system and an access panel, consistent with some embodiments of the dishwasher securing system.

[0012] FIG. **4** is a front view of a dishwasher securing system, consistent with some embodiments of the dishwasher securing system.

[0013] FIG. **5** is a front view of a dishwasher securing system, consistent with some embodiments of the dishwasher securing system.

[0014] FIG. **6** is a top view of a dishwasher securing system, consistent with some embodiments of the dishwasher securing system.

[0015] FIG. **7** is a top view of a dishwasher securing system, consistent with some embodiments of the dishwasher securing system.

[0016] FIG. **8** is a front view of a dishwasher securing system, consistent with some embodiments of the dishwasher securing system.

DETAILED DESCRIPTION

[0017] Turning now to the drawings, wherein like numbers denote like parts throughout the several views, FIG. **1** illustrates an example dishwasher and cabinet assembly **100**. The dishwasher and cabinet assembly **100** may include a cabinet set **102**. The cabinet set **102** may include a plurality of cabinet opening walls **104**. The plurality of cabinet opening walls **104** may define a cabinet opening **106**.

[0018] The dishwasher and cabinet assembly **100** includes a dishwasher **110**. The dishwasher **110** may include a housing **111** having a top face **112**, a first side face **114**, a second side face **116**, a rear face **118**, a front door **120**, and a bottom face **122**. The dishwasher **110** may be shaped and sized so as to fit within the cabinet opening **106**. A gap may exist between the dishwasher **110** and each of the cabinet opening walls **104**. In some embodiments, a fascia or trim piece may be used to cover the gap.

[0019] The housing **111** may be constructed of any suitable material, including but not limited to plastic, metals such as stainless steel or aluminum, or any other suitable material. In some embodiments, different faces of the housing **111** may be constructed of different materials, such as the top face **112** being constructed of metal and each of the side faces being constructed of plastic.

[0020] FIG. **2** illustrates a front view of a dishwasher securing system **200** configured for use on the dishwasher **110**. The dishwasher securing system **200** includes a threaded housing **210**. The threaded housing **210** includes a first sidewall **212**, a second sidewall **214**, and a top wall **215**. Each of the first sidewall **212** and second sidewall **214** are coupled to the top face **112** of the dishwasher **110**. The first sidewall **212** and second sidewall **214** may be coupled to the top face **112** of the dishwasher **110** with any suitable coupling mechanism, including but not limited to adhesives, welding, mechanical fasteners, or any other suitable coupling mechanism. In some embodiments, the threaded housing **210** may be removably coupled to the top face **112** of the dishwasher **110**.

[0021] The first sidewall **212** includes a first threaded hole **216**, and the second sidewall **214** includes a second threaded hole **218**. The first threaded hole **216** may include first internal threads **217**, and the second threaded hole **218** may include second internal threads **219**.

[0022] A first threaded leg **220** extends from the first threaded hole **216**. The first threaded leg **220** may include first external threads **221**. The first external threads **221** may be configured to threadably engage with the first threaded hole **216**.

[0023] The first threaded leg **220** may include a first end **222** and a second end **224**. The first threaded leg **220** may be configured such that when the first threaded leg **220** is threaded within the first threaded hole **216** in a first direction, the first threaded leg **220** is advanced relative to the first side face **114** of the dishwasher **110**, such that the first end **222** is an increased distance away from the first side face **114** after the first threaded leg **220** is threaded within the first threaded hole **216** in the first direction. The first threaded leg **220** may be further configured such that when the first threaded leg **220** is threaded within the first threaded hole **216** in a second direction, where the second direction is opposite the first direction, the first threaded leg **220** is retracted relative to the first side face **114** of the dishwasher **110**.

[0024] A second threaded leg **226** extends from the second threaded hole **218**. The second threaded leg **226** may include second external threads **227**. The second external threads **227** may be configured to threadably engage with the second threaded hole **218**.

[0025] The second threaded leg **226** may include a first end **228** and a second end **230**. The second threaded leg **226** may be configured such that when the second threaded leg **226** is threaded within the second threaded hole **218** in a first direction, the second threaded leg **226** is advanced relative to the second side face **116** of the dishwasher **110**, such that the first end **228** is an increased distance away from the second side face **116** after the second threaded leg **226** is threaded within the second threaded hole **218** in the first direction. The second threaded leg **226** may be further configured such that when the second threaded leg **226** is threaded within the second threaded hole **218** in a second direction, where the second direction is opposite the first direction, the second threaded leg **226** is retracted relative to the second side face **116** of the dishwasher **110**.

[0026] A first plate **240** may be coupled to the first end **222** of the first threaded leg **220**. The first plate **240** may be a plate or similar flat structure. The first plate **240** may be configured to tension against the cabinet opening walls **104** (not shown).

[0027] The first plate **240** may include an outer face **242** and an inner face **244**. A mounting bracket **246** may be coupled to the inner face **244**. The mounting bracket **246** may be flexibly coupled to the first threaded leg via a mounting pin **248**. The mounting pin **248** may allow the first plate **240** to flex relative to the first threaded leg **220** and/or the cabinet opening walls **104**.

[0028] In further embodiments, it is contemplated that the first plate **240** may be coupled to the first threaded leg **220** by any other suitable coupling mechanism, including but not limited to welding or mechanical fasteners.

[0029] A first rubber insert **250** may be coupled to the outer face **242** of the first plate **240**. The first rubber insert **250** may increase the coefficient of friction between the dishwasher securing system **200** and the cabinet set **102**. This may aid in retaining the dishwasher **110** relative to the cabinet set **102** such as when the dishwasher **110** is bumped or otherwise receives a force load.

[0030] A second plate **260** may be coupled to the first end **228** of the second threaded leg **226**. The second plate **260** may be a plate or similar flat structure. The second plate **260** may be configured to tension against the cabinet opening walls **104** (not shown).

[0031] The second plate **260** may include an outer face **262** and an inner face **264**. A mounting bracket **266** may be coupled to the inner face **264**. The mounting bracket **266** may be flexibly coupled to the second threaded leg **226** via a mounting pin **268**. The mounting pin **268** may allow the second plate **260** to flex relative to the second threaded leg **226** and/or the cabinet opening walls **104**.

[0032] In further embodiments, it is contemplated that the second plate **260** may be coupled to the second threaded leg **226** by any other suitable coupling mechanism, including but not limited to welding or mechanical fasteners.

[0033] A second rubber insert **270** may be coupled to the outer face **262** of the second plate **260**. The second rubber insert **270** may increase the coefficient of friction between the dishwasher securing system **200** and the cabinet set **102**. This may aid in retaining the dishwasher **110** relative to the cabinet set **102** such as when the dishwasher **110** is bumped or otherwise receives a force load.

[0034] FIG. **3** illustrates a top view of the dishwasher securing system **200** configured for use on the dishwasher **110**. An access panel **124** may be formed in the top face **112** of the dishwasher **110**. The access panel **124** may be a removable panel that reveals an access opening **126** in the top face **112** of the dishwasher **110**. The access opening **126** may allow an operator to access the first threaded leg **220** and the second threaded leg **226** by opening the front door **120**. This may be used such as when there is insufficient clearance between the top face **112** and the cabinet opening wall **104** for an operator to grasp the first threaded leg **220** and/or the second threaded leg **226**.

[0035] The access panel **124** may be removably coupled to the top face **112** through any suitable coupling mechanism, including but not limited to geometric fit, magnets, pins and holes, or any other suitable coupling mechanism.

[0036] FIG. **4** illustrates a front view of a dishwasher securing system **300**. The dishwasher securing system **300** includes a first threaded extension **310**. The first threaded extension **310** may include an outer end **312**, an inner end **314**, and one or more threads **316**.

[0037] The one or more threads **316** may be external threads. The dishwasher securing system **300** includes a second threaded extension **320**. The second threaded extension **320** may include an outer end **322**, and inner end **324**, and one or more threads **326**.

[0038] The dishwasher **110** may include a first threaded opening **128** and a second threaded opening **132**. The first threaded opening **128** may include one or more threads **130**. The one or more threads **130** may be internal threads. The one or more threads **130** may be configured to threadably engage with the one or more threads **316** of the first threaded extension **310**. In embodiments, the first threaded extension **310** may be advanceable outwardly from the first side face **114** of the dishwasher **110** by threading the first threaded extension **310** relative to the first threaded opening **128**.

[0039] The second threaded opening **132** may include one or more threads **134**. The one or more threads **134** may be internal threads. The one or more threads **134** may be configured to threadably engage with the one or more threads **326** of the second threaded extension **320**. In embodiments, the second threaded extension **320** may be advanceable outwardly from the second side face **116** of the dishwasher **110** by threaded the second threaded extension **320** relative to the second threaded opening **132**.

[0040] A first plate **330** may be coupled to the outer end **312** of the first threaded extension **310**. The first plate **330** may be a plate or similar flat structure. The first plate **330** may be configured to tension against the cabinet opening walls **104** (not shown).

[0041] The first plate **330** may include an outer face **332** and an inner face **334**. A mounting bracket **336** may be coupled to the inner face **334**. The mounting bracket **336** may be flexibly coupled to the first threaded leg via a mounting pin **338**. The mounting pin **338** may allow the first plate **330** to flex relative to the first threaded extension **310** and/or the cabinet opening walls **104**.

[0042] In further embodiments, it is contemplated that the first plate **330** may be coupled to the first threaded extension **310** by any other suitable coupling mechanism, including but not limited to welding or mechanical fasteners.

[0043] A first rubber insert **340** may be coupled to the outer face **332** of the first plate **330**. The first rubber insert **340** may increase the coefficient of friction between the dishwasher securing system **300** and the cabinet set **102**. This may aid in retaining the dishwasher **110** relative to the cabinet set **102** such as when the dishwasher **110** is bumped or otherwise receives a force load.

[0044] A second plate **350** may be coupled to the outer end **322** of the second threaded extension **320**. The second plate **350** may be a plate or similar flat structure. The second plate **350** may be configured to tension against the cabinet opening walls **104** (not shown).

[0045] The second plate **350** may include an outer face **352** and an inner face **354**. A mounting bracket **356** may be coupled to the inner face **354**. The mounting bracket **356** may be flexibly coupled to the second threaded extension **320** via a mounting pin **358**. The mounting pin **358** may allow the second plate **350** to flex relative to the second threaded extension **320** and/or the cabinet opening walls **104**.

[0046] In further embodiments, it is contemplated that the second plate **350** may be coupled to the second threaded extension **320** by any other suitable coupling mechanism, including but not limited to welding or mechanical fasteners.

[0047] A second rubber insert **360** may be coupled to the outer face **352** of the second plate **350**. The second rubber insert **360** may increase the coefficient of friction between the dishwasher securing system **300** and the cabinet set **102**. This may aid in retaining the dishwasher **110** relative to the cabinet set **102** such as when the dishwasher **110** is bumped or otherwise receives a force load.

[0048] FIG. **5** illustrates a front view of the dishwasher securing system **300**. In the illustrated embodiment, the dishwasher **110** may include a first mounting boss **136** and a second mounting boss **142**. The first mounting boss **136** may include a first threaded hole **138** therein. The first threaded hole **138** may include one or more threads **140**. The second mounting boss **142** may include a second threaded hole **144** therein. The second threaded hole **144** may include one or more threads **146**.

[0049] The one or more threads **140** may be configured to threadably engage with the one or more threads **316** of the first threaded extension **310**. In embodiments, the first threaded extension **310** may be advanceable outwardly from the first side face **114** of the dishwasher **110** by threading the first threaded extension **310** relative to the first threaded hole **138**.

[0050] The one or more threads **146** may be configured to threadably engage with the one or more threads **326** of the second threaded extension **320**. In embodiments, the second threaded extension **320** may be advanceable outwardly from the second side face **116** of the dishwasher **110** by threaded the second threaded extension **320** relative to the second threaded hole **144**.

[0051] The first mounting boss **136** and may be coupled to the first side face **114** by any suitable coupling mechanism, including but not limited to adhesive, magnets, mechanical fasteners, welding, geometric fit, or any other suitable coupling mechanism. The second mounting boss **142** may be coupled to the second side face **116** by any suitable coupling mechanism, including but not limited to adhesive, magnets, mechanical fasteners, welding, geometric fit, or any other suitable coupling mechanism.

[0052] FIG. **6** illustrates a top view of a dishwasher securing system **400**. The dishwasher securing system **400** includes a first rack **410** and a second rack **420**. The first rack **410** includes one or more grooved teeth **412**, an outer end **414**, and an inner end **416**. While eight grooved teeth **412** are shown, it should be understood that in embodiments, any suitable number of grooved teeth may be used, including but not limited to two grooved teeth, five grooved teeth, ten grooved teeth, or any other suitable number of grooved teeth.

[0053] The first rack **410** may be slideably coupled to the top face **112** of the dishwasher **110**. As a non-limiting example, the first rack **410** may be configured on a slider mechanism to allow the first rack **410** to slide relative to the top face **112** of the dishwasher.

[0054] The second rack **420** includes one or more grooved teeth **422**, an outer end **424**, and an inner end **426**. While eight grooved teeth **422** are shown, it should be understood that in embodiments, any suitable number of grooved teeth may be used, including but not limited to two grooved teeth, five grooved teeth, ten grooved teeth, or any other suitable number of grooved teeth.

[0055] The second rack **420** may be slideably coupled to the top face **112** of the dishwasher **110**. As a non-limiting example, the second rack **420** may be configured on a slider mechanism to allow the second rack **420** to slide relative to the top face **112** of the dishwasher.

[0056] The dishwasher securing system **400** includes a pinion **430**. The pinion **430** includes a plurality of gear teeth **432**. While thirteen gear teeth **432** are shown, it should be understood that any suitable number of gear teeth **432** may be used, including but not limited to five gear teeth, ten gear teeth, fifteen gear teeth, or any other suitable number of gear teeth.

[0057] The pinion **430** may be rotatably coupled to the top face **112** of the dishwasher **110**, such that the pinion **430** may rotate clockwise and counter clockwise relative to the top face **112** of the dishwasher **110**. In various embodiments, the pinion **430** may be rotatably coupled to the top face **112** of the dishwasher **110** via a coupling mechanism such as a bearing or an axle. However, it should be understood that in embodiments, any suitable type of coupling mechanism may be used.

[0058] The plurality of gear teeth **432** of the pinion **430** engage the plurality of grooved teeth **412** of the first rack **410** and the plurality of grooved teeth **422** of the second rack **420**. When the plurality of gear teeth **432** of the pinion are engaged with the plurality of grooved teeth **412** of the first rack **410** and the plurality of grooved teeth **422** of the second rack **420** and the pinion **430** is rotated counterclockwise, each of the first rack **410** and the second rack **420** extend outwardly relative to the first side face **114** and the second side face **116**, respectively. When the plurality of gear teeth **432** of the pinion are engaged with the plurality of grooved teeth **412** of the first rack **410** and the plurality of grooved teeth **422** of the second rack **420** and the pinion **430** is rotated clockwise, each of the first rack **410** and the second rack **420** extend inwardly relative to the first side face **114** and the second side face **116**, respectively.

[0059] It should be understood that, in embodiments, the first rack **410**, the second rack **420**, and the pinion **430** may be arranged such that clockwise rotation of the pinion **430** results in the first rack **410** and the second rack **420** extending outwardly, and counterclockwise rotation of the pinion **430** results in the first rack **410** and the second rack **420** extending inwardly.

[0060] A first plate **440** may be coupled to the outer end **414** of the first rack **410**. The first plate **440** may be a plate or similar flat structure. The first plate **440** may be configured to tension against the cabinet opening walls **104** (not shown).

[0061] The first plate **440** may include an outer face **442** and an inner face **444**. A mounting bracket **446** may be coupled to the inner face **444**. The mounting bracket **446** may be flexibly coupled to the first rack **410** via a mounting pin **448**. The mounting pin **448** may allow the first plate **440** to flex relative to the first rack **410** and/or the cabinet opening walls **104**.

[0062] In further embodiments, it is contemplated that the first plate **440** may be coupled to the first rack **410** by any other suitable coupling mechanism, including but not limited to welding or mechanical fasteners.

[0063] A first rubber insert **450** may be coupled to the outer face **442** of the first plate **440**. The first rubber insert **450** may increase the coefficient of friction between the dishwasher securing system **400** and the cabinet set **102**. This may aid in retaining the dishwasher **110** relative to the cabinet set **102** such as when the dishwasher **110** is bumped or otherwise receives a force load.

[0064] A second plate **460** may be coupled to the outer end **424** of the second rack **420**. The second plate **460** may be a plate or similar flat structure. The second plate **460** may be configured to tension against the cabinet opening walls **104** (not shown).

[0065] The second plate **460** may include an outer face **462** and an inner face **464**. A mounting bracket **466** may be coupled to the inner face **464**. The mounting bracket **466** may be flexibly coupled to the second rack **420** via a mounting pin **468**. The mounting pin **468** may allow the second plate **460** to flex relative to the second rack **420** and/or the cabinet opening walls **104**.

[0066] In further embodiments, it is contemplated that the second plate **460** may be coupled to the second rack **420** by any other suitable coupling mechanism, including but not limited to welding or mechanical fasteners.

[0067] A second rubber insert **470** may be coupled to the outer face **462** of the second plate **460**. The second rubber insert **470** may increase the coefficient of friction between the dishwasher securing system **400** and the cabinet set **102**. This may aid in retaining the dishwasher **110** relative to the cabinet set **102** such as when the dishwasher **110** is bumped or otherwise receives a force load.

[0068] FIG. **7** illustrates a top view of a dishwasher securing system **500**. The dishwasher securing system **500** may include a center pulley **510**. The center pulley **510** may be rotatably coupled to top face **112** of the dishwasher **110**, such that the center pulley **510** may rotate clockwise and counter clockwise relative to the top face **112** of the dishwasher **110**. In various embodiments, the center pulley **510** may be rotatably coupled to the top face **112** of the dishwasher **110** via a coupling mechanism such as a bearing or an axle. However, it should be understood that in embodiments, any suitable type of coupling mechanism may be used.

[0069] The center pulley **510** may have a lower groove **512** and an upper groove **514**. The lower groove **512** and the upper groove **514** may be shaped and sized to allow a cable to be disposed therein, as will be discussed in more detail herein.

[0070] The dishwasher securing system **500** may include a first extending arm **520**. The first extending arm **520** may include an inner end **522** and an outer end **524**. The first extending arm **520** may include an outer pulley **526**. The first extending arm **520** may be rotatably coupled to the top face **112** of the dishwasher **110** about the outer pulley **526**, such that the first extending arm **520** may be rotated relative to the top face **112** of the dishwasher **110** about the outer pulley **526**. The outer pulley **526** may have a groove **528** that is shaped and sized to allow a cable to be disposed therein.

[0071] The dishwasher securing system may include a first cable **530**. The first cable **530** may be a continuous loop that may be configured to be tensioned between the outer pulley **526** and the center pulley **510**. The first cable **530** may be shaped and sized to fit inside of groove **528** of the outer pulley **526** and the lower groove **512** or the upper groove **514** of the center pulley **510**. The first cable **530** may translate rotational motion between the center pulley **510** and the outer pulley **526**. This may allow the first extending arm **520** to be extended outwardly relative to the first side face **114** of the dishwasher **110** by rotation of the center pulley **510**.

[0072] The dishwasher securing system **500** may include a second extending arm **540**. The second extending arm **540** may include an inner end **542** and an outer end **544**. The second extending arm **540** may include an outer pulley **546**. The second extending arm **540** may be rotatably coupled to the top face **112** of the dishwasher **110** about the outer pulley **546**, such that the second extending arm **540** may be rotated relative to the top face **112** of the dishwasher **110** about the outer pulley

546. The outer pulley **546** may have a groove **548** that is shaped and sized to allow a cable to be disposed therein.

[0073] The dishwasher securing system **500** may include a second cable **550**. The second cable **550** may be a continuous loop that may be configured to be tensioned between the outer pulley **546** and the center pulley **510**. The second cable **550** may be shaped and sized to fit inside of groove **548** of the outer pulley **546** and the lower groove **512** or the upper groove **514** of the center pulley **510**. The second cable **550** may translate rotational motion between the center pulley **510** and the outer pulley **546**. This may allow the second extending arm **540** to be extended outwardly relative to the first side face **114** of the dishwasher **110** by rotation of the center pulley **510**.

[0074] A first plate **560** may be coupled to the outer end **524** of the first extending arm **520**. The first plate **560** may be a plate or similar flat structure. The first plate **560** may be configured to tension against the cabinet opening walls **104** (not shown).

[0075] The first plate **560** may include an outer face **562** and an inner face **564**. A mounting bracket **566** may be coupled to the inner face **564**. The mounting bracket **566** may be flexibly coupled to the first extending arm **520** via a mounting pin **568**. The mounting pin **568** may allow the first plate **560** to flex relative to the first extending arm **520** and/or the cabinet opening walls **104**.

[0076] In further embodiments, it is contemplated that the first plate **560** may be coupled to the first extending arm **520** by any other suitable coupling mechanism, including but not limited to welding or mechanical fasteners.

[0077] A first rubber insert **570** may be coupled to the outer face **562** of the first plate **560**. The first rubber insert **570** may increase the coefficient of friction between the dishwasher securing system **500** and the cabinet set **102**. This may aid in retaining the dishwasher **110** relative to the cabinet set **102** such as when the dishwasher **110** is bumped or otherwise receives a force load.

[0078] A second plate **580** may be coupled to the outer end **544** of the second extending arm **540**. The second plate **580** may be a plate or similar flat structure. The second plate **580** may be configured to tension against the cabinet opening walls **104** (not shown).

[0079] The second plate **580** may include an outer face **582** and an inner face **584**. A mounting bracket **586** may be coupled to the inner face **584**. The mounting bracket **586** may be flexibly coupled to the second extending arm **540** via a mounting pin **588**. The mounting pin **588** may allow the second plate **580** to flex relative to the second extending arm **540** and/or the cabinet opening walls **104**.

[0080] In further embodiments, it is contemplated that the second plate **580** may be coupled to the second extending arm **540** by any other suitable coupling mechanism, including but not limited to welding or mechanical fasteners.

[0081] A second rubber insert **590** may be coupled to the outer face **582** of the second plate **580**. The second rubber insert **590** may increase the coefficient of friction between the dishwasher securing system **400** and the cabinet set **102**. This may aid in retaining the dishwasher **110** relative to the cabinet set **102** such as when the dishwasher **110** is bumped or otherwise receives a force load.

[0082] FIG. **8** illustrates a front view of a dishwasher securing system **600**. The dishwasher securing system **600** may include a first pin **610** and a second pin **630**. The first pin **610** may include an inner end **612**, an outer end **614**, and a first flange **616** placed between the inner end **612** and the outer end **614**.

[0083] The first pin **610** may be disposed through a first pin hole **148** formed in the first side face **114** of the dishwasher **110**. The second pin **630** may be disposed through a second pin hole **150** formed in the second side face **116** of the dishwasher **110**.

[0084] A first spring **620** may bias the first pin **610** outwardly from the first side face **114** of the dishwasher **110**. As illustrated, the first spring **620** may be a coil spring that wraps around the first pin **610** and biases the first pin **610** by applying a spring force between the first side face **114** of the dishwasher **110** and the first flange **616** of the first pin. However, it should be understood that, in embodiments, the first spring **620** may be any suitable type of spring, including but not limited to a torsion spring, a spiral spring, a leaf spring, or any other suitable type of spring. The first flange **616** may be shaped and sized to allow an operator to compress the first spring **620** by pressing against the first flange **616**.

[0085] The second pin **630** may include an inner end **632**, an outer end **634**, and a second flange **636** placed between the inner end **632** and the outer end **634**. A second spring **640** may bias the second pin **630** outwardly from the second side face **116** of the dishwasher **110**. As illustrated, the second spring **640** may be a coil spring that wraps around the second pin **630** and biases the second pin **630** by applying a spring force between the second side face **116** of the dishwasher **110** and the second flange **636** of the second pin **630**. However, it should be understood that, in embodiments, the second spring **640** may be any suitable type of spring, including but not limited to a torsion spring, a spiral spring, a leaf spring, or any other suitable type of spring. The second flange **636** may be shaped and sized to allow an operator to compress the second spring **640** by pressing against the second flange **636**.

[0086] A first plate **650** may be coupled to the outer end **614** of the first pin **610**. The first plate **650** may be a plate or similar flat structure. The first plate **650** may be configured to tension against the cabinet opening walls **104** (not shown).

[0087] The first plate **650** may include an outer face **652** and an inner face **654**. A mounting bracket **656** may be coupled to the inner face **654**. The mounting bracket **656** may be flexibly coupled to the first pin **610** via a mounting pin **658**. The mounting pin **658** may allow the first plate **650** to flex relative to the first pin **610** and/or the cabinet opening walls **104**.

[0088] In further embodiments, it is contemplated that the first plate **650** may be coupled to the first pin **610** by any other suitable coupling mechanism, including but not limited to welding or mechanical fasteners.

[0089] A first rubber insert **660** may be coupled to the outer face **652** of the first plate **650**. The first rubber insert **660** may increase the coefficient of friction between the dishwasher securing system **600** and the cabinet set **102**. This may aid in retaining the dishwasher **110** relative to the cabinet set **102** such as when the dishwasher **110** is bumped or otherwise receives a force load.

[0090] A second plate **670** may be coupled to the outer end **634** of the second pin **630**. The second plate **670** may be a plate or similar flat structure. The second plate **670** may be configured to tension against the cabinet opening walls **104** (not shown).

[0091] The second plate **670** may include an outer face **672** and an inner face **674**. A mounting bracket **676** may be coupled to the inner face **674**. The mounting bracket **676** may be flexibly coupled to the second pin **630** via a mounting pin **678**. The mounting pin **678** may allow the second plate **670** to flex relative to the second pin **630** and/or the cabinet opening walls **104**.

[0092] In further embodiments, it is contemplated that the second plate **670** may be coupled to the second pin **630** by any other suitable coupling mechanism, including but not limited to welding or mechanical fasteners.

[0093] A second rubber insert **680** may be coupled to the outer face **672** of the second plate **670**. The second rubber insert **680** may increase the coefficient of friction between the dishwasher securing system **600** and the cabinet set **102**. This may aid in retaining the dishwasher **110** relative to the cabinet set **102** such as when the dishwasher **110** is bumped or otherwise receives a force load.

[0094] All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

[0095] The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

[0096] The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0097] As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0098] As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to that the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0099] It should also be understood that, unless clearly indicated to the contrary, in any methods claimed herein that include more than one step or act, the order of the steps or acts of the method is not necessarily limited to the order in which the steps or acts of the method are recited.

[0100] In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03.

[0101] It is to be understood that the embodiments are not limited in its application to the details of construction and the arrangement of components set forth in the description or illustrated in the drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Unless limited otherwise, the terms “connected,” “coupled,” “in communication with,” and “mounted,” and variations thereof herein are used broadly and encompass direct and indirect connections, couplings, and mountings. In addition, the terms “connected” and “coupled” and variations thereof are not restricted to physical or mechanical connections or couplings.

[0102] The foregoing description of several embodiments of the invention has been presented for purposes of illustration. It is not intended to be exhaustive or to limit the invention to the precise steps and/or forms disclosed, and obviously many modifications and variations are possible in light of the above teaching.

Claims

1. A dishwasher securing system, the system comprising: a dishwasher, the dishwasher having a housing having a top face, a bottom face, and one or more side faces; a threaded housing, the threaded housing coupled to the top face of the dishwasher, the threaded housing having a plurality of threaded holes; a first threaded leg extending from the threaded housing, the first threaded leg having one or more threads configured to engage at least one of the plurality of threaded holes of the threaded housing; and a second threaded leg extending from the threaded housing, the second threaded leg having one or more threads configured to engage at least one of the plurality of threaded holes of the threaded housing.
2. The dishwasher securing system of claim 1, wherein the top face of the dishwasher further comprises an access panel.
3. The dishwasher securing system of claim 2, wherein at least one of the first threaded leg or the second threaded leg are accessible through the access panel.
4. The dishwasher securing system of claim 1, further comprising: the first threaded leg having an outer end, wherein a first plate is coupled to the outer end of the first threaded leg; and the second threaded leg having an outer end, wherein a second plate is coupled to the outer end of the second threaded leg.
5. The dishwasher securing system of claim 4, further comprising: the first plate has an outer face, wherein a first rubber insert is coupled to the outer face of the first plate; and the second plate has an outer face, wherein a second rubber insert is coupled to the outer face of the second plate.
6. The dishwasher securing system of claim 4, wherein the first plate is flexibly coupled to the outer end of the first threaded leg, and the second plate is flexibly coupled to the outer end of the second threaded leg.
7. The dishwasher securing system of claim 1, wherein the threaded housing comprises: a first sidewall having a first threaded hole; and a second sidewall having a second threaded hole, wherein the second sidewall is opposite to the first sidewall.
8. A dishwasher securing system, the system comprising: a dishwasher, the dishwasher having a housing having a top face, a bottom face, and a plurality of side faces; and one or more threaded extensions coupled to one of the plurality of side faces, wherein each of the one or more threaded extensions are configured to be threaded outward relative to one of the plurality of side faces of the dishwasher.
9. The dishwasher securing system of claim 8, further comprising a first threaded extensions and a second threaded extension, wherein a first threaded extension is coupled to a first side face, the second threaded extension is coupled to a second side face, and the first side face and the second side face are opposite of one another.
10. The dishwasher securing system of claim 9, wherein the first side face of the housing has a first threaded hole, the second side face of the housing has a second threaded hole, wherein: the first threaded extension is configured to threadably engage the first threaded hole; and the second threaded extension is configured to threadably engage the second threaded hole.
11. The dishwasher securing system of claim 9, further comprising: the first threaded extension having an outer end, wherein a first plate is coupled to the outer end of the first threaded extension; and the second threaded extension having an outer end, wherein a second plate is coupled to the outer end of the second threaded extension.
12. The dishwasher securing system of claim 11, further comprising: the first plate has an outer face, wherein a first rubber insert is coupled to the outer face of the first plate; and the second plate

has an outer face, wherein a second rubber insert is coupled to the outer face of the second plate.
13. The dishwasher securing system of claim 11, wherein the first plate is flexibly coupled to the outer end of the first threaded extension, and the second plate is flexibly coupled to the outer end of the second threaded extension.
14. The dishwasher securing system of claim 9, further comprising: one or more mounting bosses, each of the one or more mounting bosses having a threaded hole and being mounted to one of the plurality of side faces of the dishwasher, wherein each of the one or more threaded extensions are configured to threadably engage one of the one or more mounting bosses.
15. A dishwasher securing system, the system comprising: a dishwasher, the dishwasher having a housing having a top face, a bottom face, and a plurality of side faces; a plurality of racks, each of the plurality of racks coupled to the top face of the dishwasher having a plurality of grooved teeth; one or more arms coupled to each of the plurality of racks; a pinion, the pinion having a plurality of gear teeth, wherein the pinion is configured to engage the plurality of grooved teeth of the plurality of racks, wherein: each of the plurality of racks are configured to extend outwardly when engaged by the pinion.
16. The dishwasher securing system of claim 15, further comprising: a first rack configured to be extended outwardly from a first side face of the dishwasher; and a second rack configured to be extended outwardly from a second side face of the dishwasher, wherein the second side face is opposite of the first side face.
17. The dishwasher securing system of claim 16, wherein the pinion is configured to simultaneously move the first rack and the second rack.
18. The dishwasher securing system of claim 16, wherein: the first rack further comprises an outer end, wherein a first plate is coupled to the outer end of the first rack; and the second rack further comprises an outer end, wherein a second plate is coupled to the outer end of the second rack.
19. The dishwasher securing system of claim 18, further comprising: the first plate having an outer face, wherein a first rubber insert is coupled to the outer face of the first plate; and the second plate having an outer face, wherein a second rubber insert is coupled to the outer face of the second plate.
20. The dishwasher securing system of claim 18, wherein the first plate is flexibly coupled to the outer end of the first rack, and the second plate is flexibly coupled to the outer end of the second rack.